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Who Needs to Know? Minimal Knowledge for Optimal Coordination

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Abstract

To optimally coordinate with others in cooperative games, it is often crucial to have information about one's collaborators: successful driving requires understanding which side of the road to drive on. However, not every feature of collaborators is *strategically relevant*: the fine-grained acceleration of drivers may be ignored while maintaining optimal coordination. We show that there is a well-defined dichotomy between strategically relevant and irrelevant information. Moreover, we show that, in dynamic games, this dichotomy has a compact representation that can be efficiently computed via a Bellman backup operator. We apply this algorithm to analyze the strategically relevant information for tasks in both a standard and a partially observable version of the Overcooked environment. Theoretical and empirical results show that our algorithms are significantly more efficient than baselines. Videos are available at https://minknowledge.github.io.

1. Introduction

When designing a policy for a cooperative multi-agent setting, it is often critical to have some idea of how one's co-players will behave. A policy for a self driving car must take into account various driving norms such as which side of the street other cars will drive on, and how to interpret stop light signals.

While there are many relevant features of co-player behavior that a policy designer must keep in mind, there are often many more irrelevant features that can be safely ignored. It is unnecessary for a self driving car to know the final destination of every other car, the current positions or trajectories of far away cars, or their idiosyncratic driving behaviors.

In such settings, it can be useful to separate the *strategically relevant* information from the *strategically irrelevant* information. If much of the information is irrelevant, it's easy to imagine how this could lead to significant increases in efficiency for finding optimal policies. For example, this could allow a focused effort on few-shot or zero-shot adaptation to co-players (Zand et al., 2022; Albrecht & Stone, 2017; Stone et al., 2010; Hu et al., 2020) or more efficient DecPOMDP planning algorithms (Szer & Charpillet, 2006; Seuken & Zilberstein, 2007). In order to leverage these benefits, we build the theory, data structures, and algorithms required to distinguish between relevant and irrelevant information.

We formalize the idea of the strategically relevant information via the idea of *strategic ambiguity*, described in Section 4. We show that this leads to a uniquely well-defined dichotomy between the strategically relevant and irrelevant information, captured by the *strategic equivalence relation* (SER) which defines two co-player policies as being *strategically equivalent* if and only if they have the same set of best-response policies.

We show that this strategic equivalence relation can be efficiently computed and stored in Section 5. In fact, we find that our proposed algorithm for computing these strategic equivalence relations has better computational complexity than the state of the art for finding best-response policies in DecPOMDPs.

We summarize the contributions of this paper as follows:

1. We formalize the dichotomy between strategically relevant and irrelevant information by introducing the concept of a strategic equivalence relation (SER).
2. We demonstrate and prove that SERs have a recursive substructure in dynamic games and show how they can be stored compactly in the form of a directed acyclic graph.
3. We provide novel algorithms for computing SERs in both fully-observed and partially-observed common-payoff stochastic games. These algorithms also have notable applications in more efficiently solving DecPOMDPs.
4. We demonstrate what SERs look like in various problems and use them to analyze the levels of coordination required in *Overcooked*, a popular benchmark for testing coordination in the AI literature.

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2. Related Work

There have been several lines of research towards getting AI agents to optimally coordinate in fully-cooperative multi-agent problems. Many works have explored designing multi-agent reinforcement learning methods to train agents in a centralized way to converge on Nash equilibrium in common-payoff games (Zhang et al., 2021; Foerster et al., 2016; Jin et al., 2022; Wang & Sandholm, 2002). Others have explored the setting of *ad-hoc teamwork*, in which agents aim to coordinate with teammates from a target population (Barrett & Stone, 2015; Suriadinata et al., 2021), and *zero-shot coordination*, where agents need aim to coordinate with arbitrary teammates without any prior coordination (Hu et al., 2020; Treutlein et al., 2021; Muglich et al., 2022). Our work aims to accelerate these lines of research by providing a framework that focuses on the strategically relevant aspects of the coordination problem.

A distinct line of research has been in the direction of analyzing and understanding *conventions* that agents use to overcome coordination problems (Lewis, 1975; Hadfield-Menell et al., 2019). We show that strategic equivalence relations naturally divide policies into the different conventions that agents could follow in a problem. Various measures (Fontaine et al., 2021; Fontaine & Nikolaidis, 2021) have been developed to analyze the levels of coordination required in multiagent problems. However, all of these approaches are based on simulating agents' behaviors rather than directly analyzing the problem.

Most close to our line of research are some works focused on understanding what the relevant pieces of information are in different decision making problems. Pynadath & Marsella (2007) discuss the idea of minimal models of belief that agents need to solve decision making problems. Becker et al. (2003); Witwicki & Durfee (2010); Oliehoek et al. (2021) use the insight that it is unnecessary for every agent to know everything about other agents and focus on how agents locally influence each other by exploiting the structure of particular classes of games. Several works explore settings in which games have preexisting *influence diagrams* that describe which decision variables are dependent on each other (Koller & Milch, 2003; Kearns et al., 2013; Mura, 2000). In contrast, our framework for computing strategically relevant information requires no assumptions or structure in how agents interact or influence each other within the decision making problem.

We find ourselves in the setting of cooperative (common-payoff) partially observable stochastic games, also known as decentralized partially observable Markov decision processes (DecPOMDPs). Our algorithms pull ideas from existing work on solving DecPOMDPs (Nair et al., 2003; Szer & Charpillet, 2006; Seuken & Zilberstein, 2007; Dibangoye et al., 2016). We also draw inspiration from Fagin et al. (2004) and model knowledge as *possible worlds*. That is, we represent the knowledge that agents have about co-policies through the subset of possible co-policies that they cannot tell apart.

3. Preliminaries

We consider the setting in which agents play in a common-payoff *stochastic game*, also known as a *decentralized partially observable Markov decision process* (DecPOMDP).

Definition 3.1. A *decentralized partially observable Markov decision process* is a tuple $(M, \mathcal{S}, \mathcal{A}, R, \gamma, P, \Omega, \mathcal{O})$ whose elements are defined as follows.

- $M = \{1, \dots, m\}$ is a set of agents.
- $\mathcal{S}$ is a set of states.
- $\mathcal{A} = \mathcal{A}_1 \times \cdots \times \mathcal{A}_m$ is the space of joint actions. For ease of notation, we assume without loss of generality that the action set is identical across states.
- $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ is the common reward function.
- γ is the discount factor.
- $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the transition function that satisfies $\sum_{s' \in \mathcal{S}} P(s, a, s') = 1$.
- $\Omega = \Omega_1 \times \cdots \times \Omega_m$ is the joint observation space.
- $\mathcal{O} : \mathcal{S} \times \mathcal{A} \times \Omega \to [0, 1]$ is the observation function.

A DecPOMDP is *fully-observed* if $\Omega_i = \mathcal{S}$, i.e., agents always observe the current state. We only consider *finite-horizon* DecPOMDPs with episode length T.

A stationary policy for agent i in a DecPOMDP specifies a distribution over actions $\mathcal{A}_i$ at every state for every possible history $h_i^t = [(a_i^1, o_i^1), (a_i^2, o_i^2), \dots, (a_i^t, o_i^t)]$ of play. We often interpret policies as *trees*, where the nodes are distributions over actions and the edges are observations. We say a policy is *pure* if its choice of distribution over actions is deterministic everywhere. A *Markov policy* is a policy where the action distribution depends only on the latest observation (or state). Let Π denote the joint policy space for all agents, Π_i denote the policy space of agent i, and Π_{-i} denote the *co-policy* space, the joint policy space of the co-players (all agents other than i).

Game theory. As an abuse of notation, we will sometimes write $R(\pi)$ or $R(\pi_i, \pi_{-i})$ to denote the expected return of joint policy $\pi = (\pi_i, \pi_{-i})$.

Definition 3.2. The *best-response function* for agent i is the set-valued function $\mathrm{BR}_i : \Pi_{-i} \to \mathcal{P}(\Pi_i)$ such that $\mathrm{BR}_i(\pi_{-i}) = \arg\max_{\pi_i} R(\pi_i, \pi_{-i})$, denoting the set of policies agent i could play to maximize payoff in response to the co-policy π_{-i}. The individual best-response functions for each agent can be combined to form the joint best-response function BR for all agents, where $\mathrm{BR}(\pi) = (\mathrm{BR}_1(\pi_{-1}), \mathrm{BR}_2(\pi_{-2}), \dots, \mathrm{BR}_m(\pi_{-m}))$.

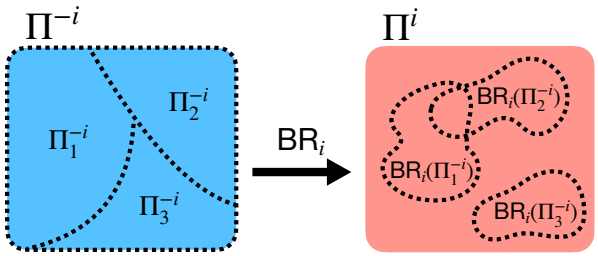

Figure 1: An abstract visualization of how the best-response function partitions the co-policy space into strategic equivalence classes. The best-response function maps the strategic equivalence classes from the co-policy space Π^{-i} (left) to their best-response subset of the policy space Π^i (right).

We call the policies $\bigcup_{\pi_{-i} \in \Pi_{-i}} \mathrm{BR}(\pi_{-i})$, the set of *best-response policies* for player i.

In our context, a joint policy π is a *subgame perfect equilibrium* if, for all states $s \in S$,

$$Q^\pi(s, \pi(s)) \geq Q^\pi(s, a_i, \pi_{-i}(s)), \quad \forall a_i \in \mathcal{A}_i, \quad (1)$$

for all agents $i \in [m]$, where Q^π is the Q-function induced by following the policy π in the future.

4. Strategic relevance

4.1. The minimal knowledge required to coordinate

Consider an agent, Alice, driving on the road. Alice feels safe driving on the road because she knows that the drivers around her will follow certain conventions and rules. She knows that other drivers will stop at red lights and go at green lights, that everyone will drive on the right side of the road, and that everyone will alternate who gets to go at stop signs. All of these conventions give Alice knowledge about other driver's policies that is relevant to how she can successfully coordinate with them on the open road. Despite this knowledge that Alice has about other drivers' behaviors, many aspects, if not most, are unknown to Alice. She does not know what the other drivers' destinations are, whether the other cars are manual or automatic, or the fine-grained motor controls of drivers. Luckily, from Alice's perspective, these other details about other drivers' behavior are irrelevant to her task of safely driving on the road.

In the rest of this section, we formalize the question: how can we distinguish between the knowledge of other agents' policies that is relevant and irrelevant to the task?

4.2. Strategic ambiguity

Let $\Pi'_{-i} \subset \Pi_{-i}$ be the subset of policies that agent i thinks their co-policy could possibly be, given their current knowledge. If agent i can narrow down Π'_{-i} to a small enough subset, then agent i has enough knowledge to exactly characterize what their optimal response should be. In such settings, we call Π'_{-i} *strategically unambiguous*.

Definition 4.1. A nonempty subset $\tilde{\Pi}_{-i} \subset \Pi_{-i}$ of policies is *strategically unambiguous* if the best-response set to all elements of $\tilde{\Pi}_{-i}$ are identical: $\mathrm{BR}_i(\pi_{-i}) = \mathrm{BR}_i(\pi'_{-i})$ for all $\pi_{-i}, \pi'_{-i} \in \tilde{\Pi}_{-i}$. We call sets of policies that don't satisfy this property *strategically ambiguous*.

If Π'_{-i} is strategically ambiguous, it means that agent i does not have enough information to unambiguously know what their best response is; they are lacking some strategically relevant information. Let's consider the two extremes. If agent i knows agents $-i$'s policies exactly, then the policies Π'_{-i} that agent i thinks their co-players are following is a singleton and is always trivially strategically unambiguous. On the other hand, if agent i knows nothing, then $\Pi'_{-i} = \Pi_{-i}$ is the full policy space. However, in a fully decentralized problems where agent i does not not need to know anything about their co-policy, the full policy space Π_{-i} is still strategically unambiguous.

A natural next question to ask is: what is the *minimal* amount of knowledge an agent needs to have about their co-policy to characterize what policies are optimal? Or dually, what is the *maximal* subset of policies that is still strategically unambiguous?

Definition 4.2. A strategically unambiguous subset $\Pi'_{-i} \subset \Pi_{-i}$ of policies *contains no irrelevant information* if any strict superset of Π'_{-i} is strategically ambiguous.

A strict superset of Π'_{-i} corresponds to a scenario in which agent i has less knowledge. So a subset that contains no irrelevant information corresponds with a scenario in which *any* less information would lead to strategic ambiguity, In other words, all of the strategically irrelevant information has already been discarded.

4.3. Strategic equivalence relations

Now we turn our attention to partitioning an agent's co-policy space in terms of best responses, and show how this relates to strategic ambiguity.

All of the strategically unambiguous subsets of Π_{-i} that contain no irrelevant information are given by taking the preimage of the best response function for agent i.

Theorem 1. The strategically unambiguous subsets of Π_{-i} that contain no irrelevant information are given by the preimage of the best-response map $\mathrm{BR}_i : \Pi_{-i} \to \mathcal{P}(\Pi_i)$.

Proof. We will show that a strategically unambiguous subsets of Π_{-i} contain no irrelevant information if and only if it is in the preimage of the best-response map. Consider $\mathrm{BR}_i^{-1} : \mathcal{P}(\Pi_i) \to \Pi_{-i}$, the inverse of BR_i.